Name: Jose Salinas
Course: CS-6370
Professor: Meilin Liu
Date: 9/15/2026
Project Name: Project 1 MatrixAdd MatrixMult

Report:

Both matrix addition and multiplication program compile. All project files have course number, professor name, data, and the project name. The README will instruct how to compile and use the Matrix addition and multiplication program. Executing the program will execute respective calculations on CPU and GPU and print the results on the screen. Both multiplication and addition executions results are correct. For small matrices (8 x 8) both will print the whole matrix. Anything larger than 8x8 it will print the first 20 elements of the matrix.

The Matrix Addition program will add two 2D matrices on the device GPU and CPU, while the matrix multiplication program will multiply both 2D matrices. The matrices that are calculated in both programs will be the same on the GPU as in the CPU. The GPU results will be transferred back to to the CPU. Both programs will compare the GPU-computed matrix with the CPU computed matrix and will print “Test PASSED” if they match.

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MatrixAdd Screenshots:

```
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixAdd
Enter Thread Block Size: 4
Enter Matrix Size: 8
THREAD_BLOCK_SIZE: 4 x 4
MATRIX_SIZE: 8 x 8
GRID_SIZE: x: 2 y: 2
All Matrix Elements in GPU Matrix
854 284 746 838 761 325 353 801
525 889 697 203 752 105 527 402
413 649 673 846 573 653 883 895
291 337 445 453 -3 287 105 965
841 180 697 793 253 691 509 928
837 495 744 185 112 181 779 467
594 966 816 766 801 984 688 899
796 365 684 855 550 328 6 451
All Matrix Elements in CPU Matrix
854 284 746 838 761 325 353 801
525 889 697 203 752 105 527 402
413 649 673 846 573 653 883 895
291 337 445 453 -3 287 105 965
841 180 697 793 253 691 509 928
837 495 744 185 112 181 779 467
594 966 816 766 801 984 688 899
796 365 684 855 550 328 6 451
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixAdd
Enter Thread Block Size: 16
Enter Matrix Size: 128
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 128 x 128
GRID_SIZE: x: 8 y: 8
First 20 elements in GPU Matrix
854 284 746 838 761 325 353 801 525 889 697 203 752 105 527 402 413 649 673 846
First 20 elements in CPU Matrix
854 284 746 838 761 325 353 801 525 889 697 203 752 105 527 402 413 649 673 846
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixAdd
Enter Thread Block Size: 16
Enter Matrix Size: 500
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 500 x 500
GRID_SIZE: x: 32 y: 32
First 20 elements in GPU Matrix
854 284 746 838 761 325 353 801 525 889 697 203 752 105 527 402 413 649 673 846
First 20 elements in CPU Matrix
854 284 746 838 761 325 353 801 525 889 697 203 752 105 527 402 413 649 673 846
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixAdd
Enter Thread Block Size: 16
Enter Matrix Size: 1000
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 1000 x 1000
GRID_SIZE: x: 63 y: 63
First 20 elements in GPU Matrix
854 284 746 838 761 325 353 801 525 889 697 203 752 105 527 402 413 649 673 846
First 20 elements in CPU Matrix
854 284 746 838 761 325 353 801 525 889 697 203 752 105 527 402 413 649 673 846
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$
```

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MatrixMult Screenshots:

```
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixMult
Enter Thread Block Size: 4
Enter Matrix Size: 8
THREAD_BLOCK_SIZE: 4 x 4
MATRIX_SIZE: 8 x 8
GRID_SIZE: x: 2 y: 2
GPU time taken 0 seconds 0 milliseconds
CPU time taken 0 seconds 0 milliseconds
All Matrix Elements in GPU Matrix
-5566 -2222 -4542 -4830 -4670 -1438 -638 -2774
-2932 500 -2468 -7596 -3100 -4372 1692 -5740
-1615 4669 457 -2027 1649 -771 185 -1627
3816 6952 5672 3048 6040 1256 5880 4040
-1980 1028 -1428 52 2852 524 -1740 -3940
-2872 2336 -920 -2944 -1872 8 40 528
1602 642 1266 -1278 -630 -1030 1730 882
1035 767 1467 103 75 -1625 2643 2047
All Matrix Elements in CPU Matrix
-5566 -2222 -4542 -4830 -4670 -1438 -638 -2774
-2932 500 -2468 -7596 -3100 -4372 1692 -5740
-1615 4669 457 -2027 1649 -771 185 -1627
3816 6952 5672 3048 6040 1256 5880 4040
-1980 1028 -1428 52 2852 524 -1740 -3940
-2872 2336 -920 -2944 -1872 8 40 528
1602 642 1266 -1278 -630 -1030 1730 882
1035 767 1467 103 75 -1625 2643 2047
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixMult
Enter Thread Block Size: 16
Enter Matrix Size: 128
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 128 x 128
GRID_SIZE: x: 8 y: 8
GPU time taken 0 seconds 0 milliseconds
CPU time taken 0 seconds 6 milliseconds
First 20 elements in GPU Matrix
-25904 -16632 -17992 -14784 -11912 -23264 -10392 -17472 -23008 -21960 -28592 -17768 -23008 -10192 -28736 -1
96 -12024 -17168 -18960 -7952
First 20 elements in CPU Matrix
-25904 -16632 -17992 -14784 -11912 -23264 -10392 -17472 -23008 -21960 -28592 -17768 -23008 -10192 -28736 -1
96 -12024 -17168 -18960 -7952
Test PASSED
```

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```
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixMult
Enter Thread Block Size: 16
Enter Matrix Size: 500
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 500 x 500
GRID_SIZE: x: 32 y: 32
GPU time taken 0 seconds 1 milliseconds
CPU time taken 0 seconds 368 milliseconds
First 20 elements in GPU Matrix
-85292 -45508 -43500 -32532 -51828 -42588 -66996 -62852 -31404 -52492 -36068 -41772 -55900 -66772 -48508 -3
72 -56332 -67884 -79732 -18532
First 20 elements in CPU Matrix
-85292 -45508 -43500 -32532 -51828 -42588 -66996 -62852 -31404 -52492 -36068 -41772 -55900 -66772 -48508 -3
72 -56332 -67884 -79732 -18532
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$ ./matrixMult
Enter Thread Block Size: 16
Enter Matrix Size: 1024
THREAD_BLOCK_SIZE: 16 x 16
MATRIX_SIZE: 1024 x 1024
GRID_SIZE: x: 64 y: 64
GPU time taken 0 seconds 9 milliseconds
CPU time taken 4 seconds 257 milliseconds
First 20 elements in GPU Matrix
-31327 -563 -62671 -37611 -45111 -49243 -21719 -77083 -7183 -34491 -26927 -45707 -42679 -56283 -15607 -462
-59439 -63611 -27623 -36915
First 20 elements in CPU Matrix
-31327 -563 -62671 -37611 -45111 -49243 -21719 -77083 -7183 -34491 -26927 -45707 -42679 -56283 -15607 -462
-59439 -63611 -27623 -36915
Test PASSED
(base) JoseLaptop:[salinaj2]:~/Dev/LearningRepo/CUDA_Parallel_Programming/vecAdd$
```